

Comp683: Computational Biology

Lecture 18

April 8, 2026

Today

Announcements:

- homework 2 is due April 15.
- Journal club 2, Monday April 20. [https://www.cell.com/immunity/fulltext/S1074-7613\(25\)00433-9](https://www.cell.com/immunity/fulltext/S1074-7613(25)00433-9)

Today:

- Begin multi-modal integration for biomedical datasets
- Specifying a *joint subspace* for multiple samples across several modalities
- Linear algebra tricks - Rayleigh Ritz Theorem

Notation and Problem Formulation

- Consider M types of omics data measurements $\{\mathbf{X}^m\}_{m=1}^M$ from the same set of N patients.
- For a modality, m , there are p_m measured features and the dimensions of the data matrix are therefore $p_m \times N$
- We will let G^m be the graph for modality m
- **Goal:** We seek a joint subspace embedding, $\mathbf{U} \in \mathbb{R}^{N \times k}$ that is representative of all modalities.

Comment

Before we had node2vec, we just used nice theorems from linear algebra!
:D (the OG graph embedding)

Overview of Subspace Merging

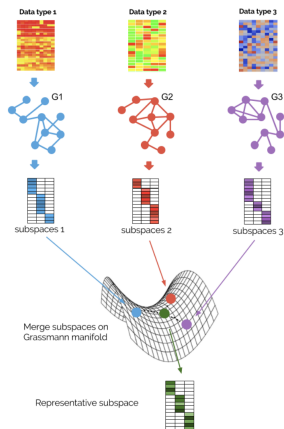


Figure: from Ding *et al.* Bioinformatics. 2019.

What is a Grassmann Manifold?

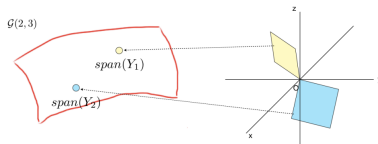


Figure: Example of $\mathcal{G}(2, 3)$

- Notation, $\mathcal{G}(k, n)$ is the set of k -dimensional linear subspaces in $\mathbb{R}^n$, such that each subspace is a point.
- So, each point on $\mathcal{G}$ can be represented by an orthonormal matrix $\mathbf{Y} \in \mathbb{R}^{n \times k}$
- **Selling Point:** We know how to talk about how geometrically close the subspaces are, based on principle angles

Why is this useful to our problem?

- **Each Modality Graph As A Subspace:** From each modality, we create a graph. We can ultimately compute the joint subspace or *embedding* given individual subspaces.
- **Well-Defined Distances Measures:** We know how to compute distances between subspaces on the Grassmannian. The representative subspace, $\mathbf{U}$ should be equidistant from the per-modality subspaces ($\mathbf{U}^m$ s).

Build a Similarity Graph Between Patients in Each Modality

Use our 'favorite' rule for calculating edge weights as,

$$S_{ij}^m = \exp \left(-\frac{\|\mathbf{x}_i^m - \mathbf{x}_j^m\|^2}{2t^2} \right), i = 1, \dots, N, j = 1, \dots, N$$

From here, retain the top k edges for each node based on S_{ij} and use W_{ij} for the notation of the edge weights retained, such that, $W_{ij}^m = S_{ij}^m$

Quadratic form helps with optimization problem

We already talked about the total variation of a signal in terms of the Graph Laplacian, or the variation of a signal around neighbors as,

$$\mathbf{x}^T \mathbf{L} \mathbf{x} = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N A_{ij} (x_i - x_j)^2 \quad (1)$$

This is useful for us, because you can think of $\mathbf{x}$ as a dimension of the embedding coordinates, which should not vary too much across the graph.

Pause for Rayleigh Ritz Theorem

Let $\mathbf{A}$ be a square, symmetric matrix, $N \times N$ matrix with eigenvalues, $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$ and corresponding eigenvectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. Then define

$$R_{\mathbf{A}}(\mathbf{x}) = \frac{\mathbf{x}^T \mathbf{A} \mathbf{x}}{\mathbf{x}^T \mathbf{x}}. \quad (2)$$

Then the minimum value of $R_{\mathbf{A}}(\mathbf{x})$ is λ_1 and it's taken for $\mathbf{x} = \mathbf{v}_1$

Matrix Extension

We will be seeing a lot on the form of $\mathbf{X}^T \mathbf{L} \mathbf{X}$. We can talk about the trace of that matrix product as the distance in vectors of adjacent nodes.

$$\text{trace}(\mathbf{X}^T \mathbf{L} \mathbf{X}) = \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N A_{ij} \|\mathbf{x}_i - \mathbf{x}_j\| \quad (3)$$

An extension of Rayleigh Ritz says that the minimum k -dimensional matrix $\mathbf{X}$ of $\text{trace}(\mathbf{X}^T \mathbf{L} \mathbf{X})$ is $\lambda_1 + \lambda_2 + \dots + \lambda_k$ and is obtained using the first k eigenvectors of $\mathbf{L}$, as $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$.

Specify Optimization Problem in terms of Normalized Graph Laplacian

$$\mathbf{L}^m = \mathbf{D}^{m-\frac{1}{2}} (\mathbf{D}^m - \mathbf{W}^m) \mathbf{D}^{m-\frac{1}{2}}$$

Written out this gives us,

$$L_{i,j}^{\text{sym}} := \begin{cases} 1 & \text{if } i = j \text{ and } \deg(v_i) \neq 0 \\ -\frac{1}{\sqrt{\deg(v_i) \deg(v_j)}} & \text{if } i \neq j \text{ and } v_i \text{ is adjacent to } v_j \\ 0 & \text{otherwise.} \end{cases}$$

Writing Down the Objective Function

The goal is to specify a $\mathbf{U}^m$ for each modality. The optimal graph embedding in k dimensions can be written as,

$$\min_{\mathbf{U}^m \in \mathbb{R}^{N \times k}} \text{tr} \left(\mathbf{U}^{m'} \mathbf{L}^m \mathbf{U}^m \right), \quad \text{s.t. } \mathbf{U}^{m'} \mathbf{U}^m = \mathbf{I}$$

- It turns out the solution is the first k eigenvectors of the Graph Laplacian $\mathbf{L}^m$ by the Rayleigh–Ritz theorem

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¹Note that the $\mathbf{U}^{m'}$ refers to the transpose of $\mathbf{U}^m$

Defining a Projection Distance Between The Integrative Subspace and Individual Modality Subspaces

$$\begin{aligned}d_{\text{proj}}^2 \left(\mathbf{U}, \{\mathbf{U}^m\}_{m=1}^M \right) &= \sum_{m=1}^M d_{\text{proj}}^2 (\mathbf{U}, \mathbf{U}^m) \\&= \sum_{m=1}^M [k - \text{tr} (\mathbf{U}\mathbf{U}'\mathbf{U}^m\mathbf{U}^{m'})] \\&= kM - \sum_{i=1}^M \text{tr} (\mathbf{U}\mathbf{U}'\mathbf{U}^m\mathbf{U}^{m'})\end{aligned}$$

The subspace, $\mathbf{U}$ that minimizes this is close to all individual subspaces, $\{\mathbf{U}^m\}_{i=1}^M$

Optimization Problem for Multiple Subspaces

The optimization problem for merging multiple subspaces finally can be written as,

$$\min_{\mathbf{U} \in \mathbb{R}^{n \times k}} \sum_{m=1}^M \text{tr}(\mathbf{U}' \mathbf{L}^m \mathbf{U}) + \alpha \left[kM - \sum_{m=1}^M \text{tr}(\mathbf{U} \mathbf{U}' \mathbf{U}^m \mathbf{U}^{m'}) \right], \quad \text{s.t. } \mathbf{U}' \mathbf{U} = \mathbf{I}$$

The authors showed that this simplifies to,

$$\min_{\mathbf{U} \in \mathbb{R}^{n \times k}} \text{tr} \left[\mathbf{U}' \left(\sum_{i=1}^M \mathbf{L}^i - \alpha \sum_{m=1}^M \mathbf{U}^m \mathbf{U}^{m'} \right) \mathbf{U} \right], \quad \text{s.t. } \mathbf{U}' \mathbf{U} = \mathbf{I}$$

Rayleigh Ritz Again....

$$\min_{\mathbf{U} \in \mathbb{R}^{n \times k}} \text{tr} \left[\mathbf{U}' \left(\sum_{i=1}^M \mathbf{L}^m - \alpha \sum_{m=1}^M \mathbf{U}^m \mathbf{U}^{m'} \right) \mathbf{U} \right], \quad \text{s.t. } \mathbf{U}'\mathbf{U} = \mathbf{I}$$

Hopefully you recognize the form of the objective. We can define a new matrix, $\mathbf{L}_{\text{mod}}$ and again the first k eigenvectors are the optimal solution. Or,

$$\mathbf{L}_{\text{mod}} = \sum_{m=1}^M \mathbf{L}^m - \alpha \sum_{m=1}^M \mathbf{U}^m \mathbf{U}^{m'}$$

Recap

- ① **Per-Modality Graph:** Create a graph for each modality and compute corresponding Laplacians. These form the points on the Grassmannian.
- ② **Quadratic Form for Per-Modality Subspaces :** In general we want to ensure each subspace dimension respects each modality's graph structure and hence yields a small value for the quadratic form.
- ③ **Projection Distance:** The ultimate joint subspace, $\mathbf{U}$ should be as close as possible to per-modality subspaces, $\mathbf{U}^m$ s
- ④ **Apply Rayleigh Ritz:** Objective is formulated in a way that we know the optimal solution is the first k eigenvectors.

Clustering on Merged Subspace

When you cluster on the merged subspace, you get groups with different prognostic interpretations.

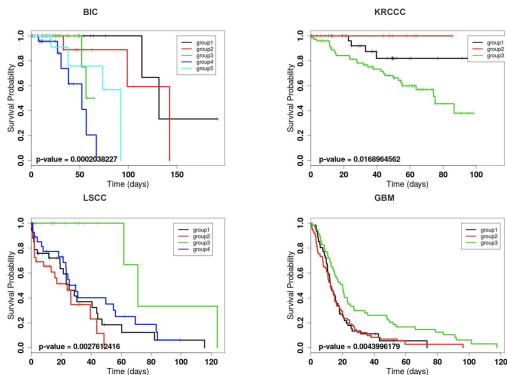


Figure: from Ding *et al.* Bioinformatics. 2018.

Another View : Between Patient Similarity

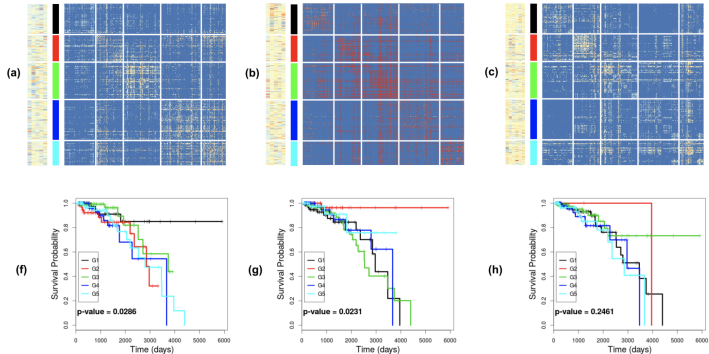


Figure: from Ding *et al.* Bioinformatics. 2018. Here we are viewing adjacency matrices between patients, based on all features jointly.

For Uncovering Trajectories Based on scRNA-seq Data

The joint subspace was used to infer the trajectory or ordering of cells.

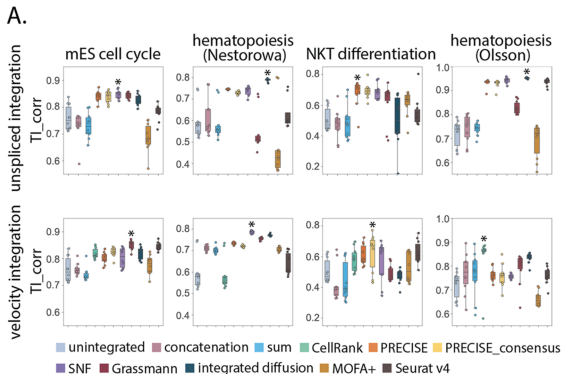


Figure: from Ranek *et al.* Genome Biology. 2022. Grassmann does pretty well, especially for integrating RNA velocity information.

Integrating Heterogeneous Information Sources

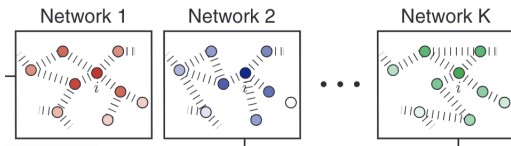


Figure: from Cho *et al.* Cell Systems. Each graph is representing a different relational definition between features.

Considering proteins, there are multiple methods for predicting whether these proteins interact .

- Physical binding
- gene expression
- co-localization
- experimentally determined
- text mined, etc.

We Seek a Unified Representations of these Nodes

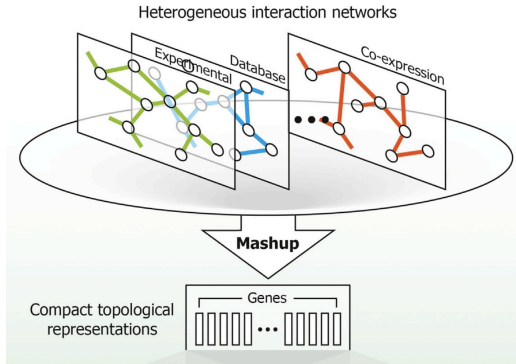


Figure: from Cho *et al.* Cell Systems. 2016.

Example from STRING

Using the STRING database, you can extract PPIs according to multiple relational definitions.

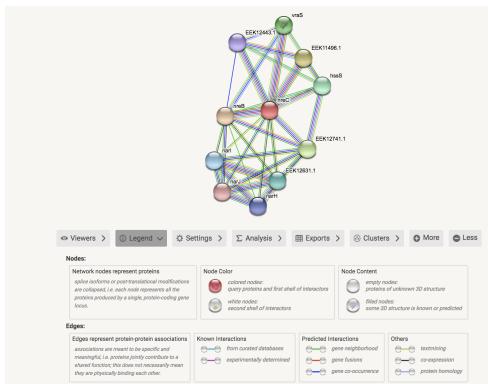


Figure: <https://string-db.org/>

Welcome Mashup

Given multiple relational definitions (e.g. multiple graphs) between a common set of nodes (features), define a consensus d -dimensional embedding vector for each node that aligns well with each individual graph (e.g. distinct relational definitions).

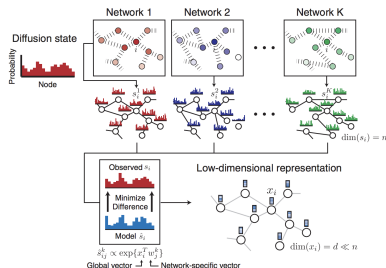


Figure: from Cho *et al.* Cell Systems. Each graph is representing a different relational definition between node (features).

Random Walk with Restart

- RWR is a way to account for both local and global 'walk' information in the graph by giving your walker the chance to restart

But first, let's re-define the transition probability that a walker goes from node j to node i as,

$$B_{ij} = \frac{A_{ij}}{\sum_{i'} A_{i'j}}$$

RWR Formally Written

Given the transition matrix, B , the RWR from a node i is defined as,

$$s_i^{t+1} = (1 - p_r)Bs_i^t + p_re_i$$

- p_r is the probability of restart
- e_i is an n -dimensional vector with $e_i(i) = 1$ and $e_i(j) = 0$ for $j \neq i$
- s_i^t is the vector of probabilities of each node being visited after t steps in the random walk, starting from node i

Clarifying What is Happening Here

$$s_i^{t+1} = (1 - p_r)Bs_i^t + p_re_i$$

- The first term corresponds to following a random edge connected to the current node
- The second term corresponds to restarting from node i .
- At some point, this reaches a stationary distribution, s_i^∞ , or fixed point
- When the diffusion states between two nodes are close, this implies they have similar positions in the graph with respect to other nodes.

Quantifying Topological Overlap Between a Node Pair

Each node is given two vector representations, $\mathbf{w}_i, \mathbf{x}_i \in \mathbb{R}^d$

- Let $\mathbf{w}_i$ refer to the context feature of a node (e.g. per relational definition)
- Let $\mathbf{x}_i$ refer to the node feature of node i (e.g. overall)

Define a new similarity measure between nodes i and j as,

$$\hat{s}_{ij} = \frac{\exp\{\mathbf{x}_i^T \mathbf{w}_j\}}{\sum_{j'} \exp\{\mathbf{x}_i^T \mathbf{w}_{j'}\}}$$

Unpacking

$$\hat{s}_{ij} = \frac{\exp\{\mathbf{x}_i^T \mathbf{w}_j\}}{\sum_{j'} \exp\{\mathbf{x}_i^T \mathbf{w}_{j'}\}}$$

- If $\mathbf{x}_i$ and $\mathbf{w}_j$ are close in direction and hence have a large inner product, then node j should be frequently visited in the random walk starting from node i .

Recap of what is happening

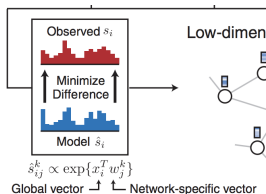


Figure: from Fig. 1. Given observed diffusion states from RWR, we should be able to find a global vector ($\mathbf{x}$) and view-specific vector ($\mathbf{w}$), such that a function of $\mathbf{x}$ and $\mathbf{w}$ gives a good diffusion state approximation.